

• The service and maintenance infrastructure is still underdeveloped compared to LPG networks

Safety considerations cannot be overlooked. Hydrogen can burn across a wide concentration range (approximately 4 to 75 per cent by volume in air) and produces a nearly invisible, rapidly propagating flame. As a result, gas-flow control, rapid leak detection, adequate ventilation, and effective flame monitoring become critical safety requirements. Compliance with guidelines established by the Bureau of Indian Standards and the International Organisation for Standardisation is essential for safe domestic deployment.

Adoption is likely to occur gradually, beginning with commercial kitchens and premium urban households before expanding as costs decline and supporting infrastructure matures.

Where hydrogen stands today

Globally, hydrogen technologies are being developed by companies such as Toyota and Panasonic. Japan, for example, has deployed household hydrogen-based energy systems such as ENE-FARM, which generate electricity and heat for homes but are not designed for cooking applications. Across Europe, decentralised hydrogen production is being explored, although most hydrogen deployments remain concentrated in industrial and transportation sectors.

The physics of one meal—What powers your *roti*?

To prepare a simple Indian meal (2-3 *rotis* and *sabzi*), a hydrogen stove functions like a miniature energy:

- ~0.8kWh-1kWh electricity consumed
- ~80ml-120ml of water is split into hydrogen and oxygen
- Electrolysis efficiency: ~70%
- Hydrogen burned instantly for flame-based cooking

What this really means:

- The true fuel is not water but electricity, temporarily stored as hydrogen
- The flame you see is energy converted twice: electricity→hydrogen→heat
- Once the stove is switched off, fuel production stops—there is no stored fuel and no residue

In one sentence: A hydrogen stove does not store energy; it converts and uses it in real time.

The key innovation in hydrogen cooking lies in producing and consuming fuel within the same appliance. However, in most urban environments, hydrogen cooking is unlikely to replace conventional cooking technologies in the near term due to higher costs and lower overall efficiency than electrified appliances, particularly induction cookers.

Future developments could see hydrogen stoves integrated into broader household energy systems. Electricity generated by rooftop solar installations could be used to produce hydrogen through electrolysis, which could then be used for cooking or converted back into electricity using fuel cells.

Hydrogen cooking is therefore more likely to remain focused on specialised markets, such as off-grid homes, clean-energy households, and premium segments, rather than replacing LPG and induction cooktops outright. Progress in

hydrogen cooking innovation will largely depend on three factors: affordability, access to clean energy, and robust engineering design.

More broadly, hydrogen stoves point towards a fundamental shift in how homes meet their energy needs. Rather than functioning solely as consumers of energy, households could increasingly generate, store, convert, and utilise energy through technologies such as electrolysis and rooftop photovoltaic systems. In the long term, homes could evolve from passive energy consumers into active producers and storage hubs. In this sense, hydrogen stoves represent more than innovative kitchen appliances; they form part of a wider decentralised household energy ecosystem. **EFY**

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